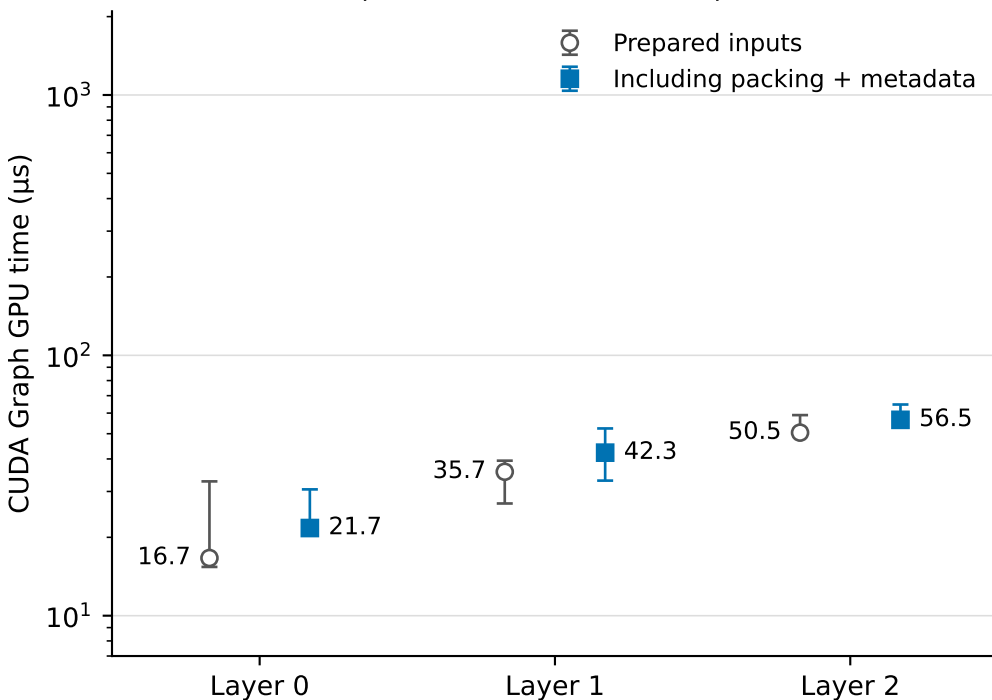
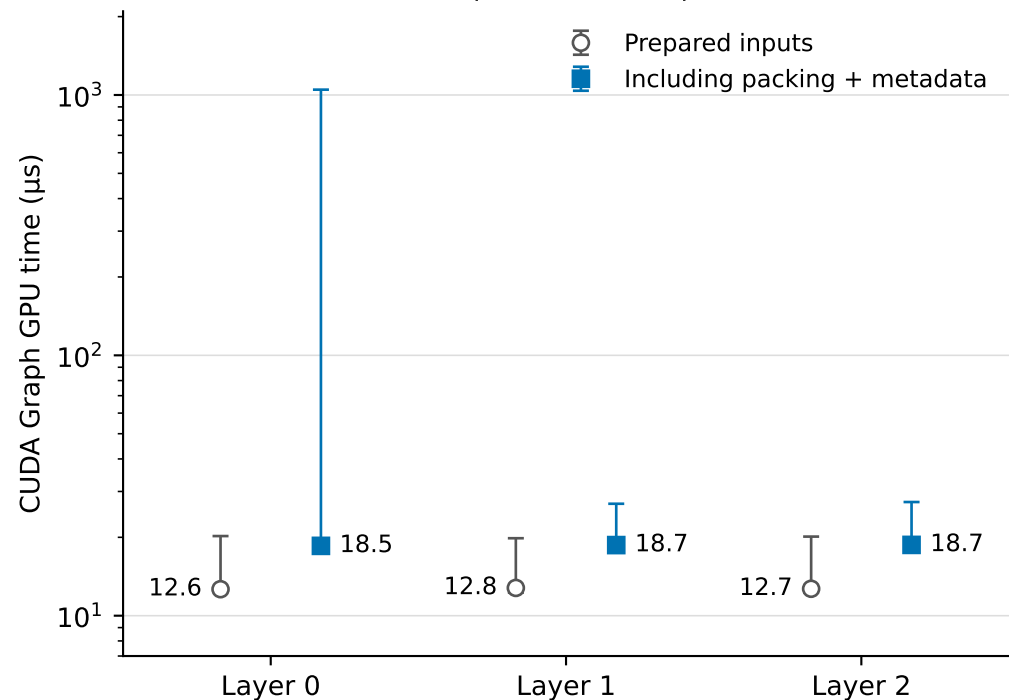


Official ECHO Q1 source bridge · N = 65,537, cap = 64

Cold map, threshold 0: 64 records prefetched



Resident map: zero records prefetched



GPU1 · log scale · median of 30 single replays; whiskers: full sample range, including outliers.
Exact top-k, staging promotion and residual recall excluded; resident map is a raw-kernel condition.